Alert Triage with Threat Intelligence

Objective

The objective of this task was to perform alert triage using the Wazuh SIEM platform and validate indicators of compromise (IOCs) through external threat intelligence sources such as VirusTotal and AlienVault OTX. This process helps SOC analysts determine whether suspicious activity represents a real security threat or simulated detection behavior.

Tools Used

The following tools were used to complete this task:

- Wazuh SIEM platform for alert monitoring and detection
- VirusTotal threat intelligence platform for IOC validation
- AlienVault OTX threat intelligence feed for reputation lookup
- Linux syslog simulation for alert generation

Alert Triage Simulation

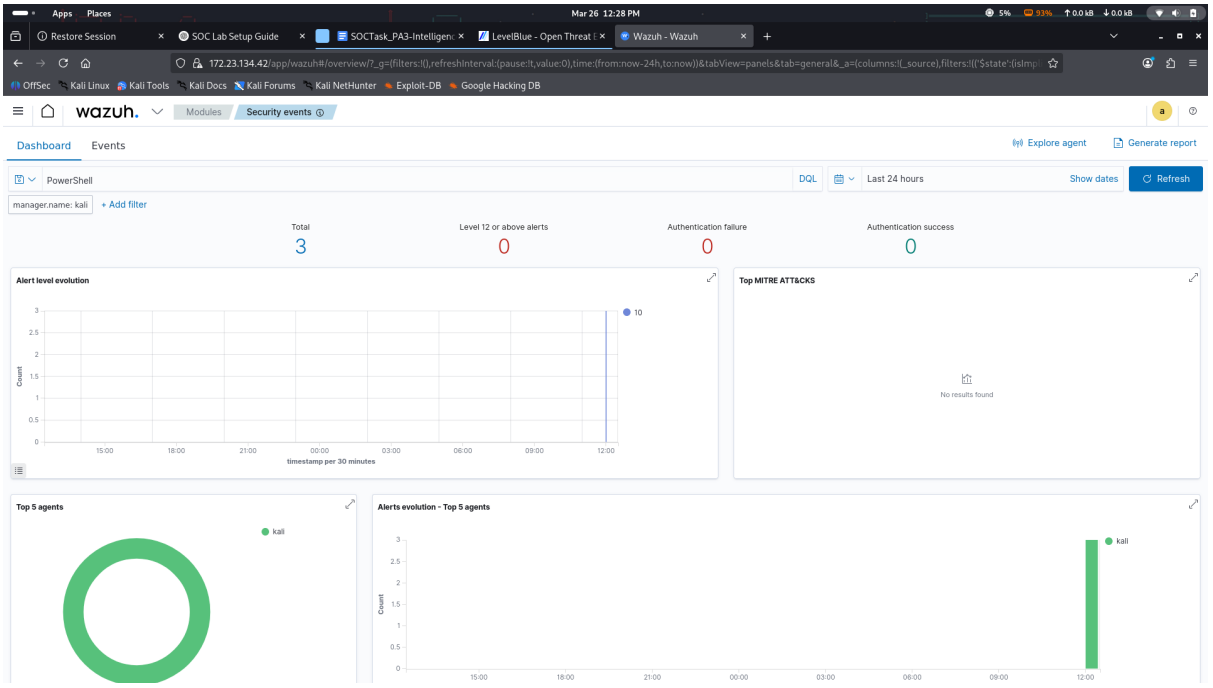
A simulated suspicious activity alert representing PowerShell execution behavior was generated inside the Wazuh dashboard. This alert represents a common attacker technique where PowerShell is used for script execution, payload delivery, or privilege escalation.

The alert was analyzed inside the Wazuh Security Events dashboard to identify its severity level, source IP address, and associated detection rule description.

The details of the alert are shown below:

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

The alert was categorized as High priority because suspicious PowerShell execution is frequently associated with attacker activity such as remote command execution, malware staging, or credential harvesting attempts.



IOC Validation using Threat Intelligence Platforms

The indicator of compromise associated with the alert was the source IP address:

192.168.1.101

This IP address was investigated using both VirusTotal and AlienVault OTX threat intelligence platforms to determine whether it had any known malicious reputation.

The lookup results showed that the IP address belongs to a private internal network range and does not appear in public threat intelligence databases. This confirms that the alert represents simulated activity used to validate SIEM detection and alert triage workflows rather than a confirmed external threat source.

Despite being non-malicious, the investigation demonstrates how SOC analysts validate indicators using external intelligence sources before escalating alerts.

IOC Validation Summary

The suspicious PowerShell execution alert generated in Wazuh was investigated using VirusTotal and AlienVault OTX threat intelligence platforms. The associated source IP address did not match known malicious indicators. This confirms the alert represents simulated detection activity and demonstrates successful alert triage and IOC validation workflow implementation within the SOC environment.

Conclusion

This task demonstrated the process of alert triage using the Wazuh SIEM platform and external threat intelligence services. The suspicious PowerShell execution alert was analyzed and validated using VirusTotal and AlienVault OTX to determine its reputation status. The activity confirmed simulated behavior while reinforcing practical SOC investigation procedures for indicator validation and alert prioritization.